Week :10 (2-6 November 2026)

من – إلى الأسبوع #:

Subject/الموضوع: Physics **Unit Title/الوحدة عنوان: Standing waves and Structure of an atom** **Grade/: 12**

Lessons الحصص	Learning targets أهداف التعلم	Assessment tools and strategies التقييم Formativeتكويني/ Summativeختامي / Differentiation التمايز	Resources and links for Parents و المصادر الروابط المخصصة لأولياء الأمور
Lesson 1: الحصة 1 قانون بويل	I can explain the Doppler effect and describe how the relative motion between a source and an observer affects the observed frequency.	2.2 Clarify syntax/structure and highlight relationships in text, formulas, diagrams. 2.1 Clarify vocabulary and symbols using visuals, connections, and supports.	please check student Teams
Lesson 2 الحصة الثانية	I can explain how standing waves form and identify their key features.	1.1 Flexible display of information (size, color, font, layout). 2.1 Clarify vocabulary and symbols using visuals, connections, and supports.	please check student Teams
Lesson 3 الحصة الثالثة	I can interpret nuclear notation and explain how emission and absorption spectra provide evidence for discrete atomic energy levels.	2.5 Use multiple media to illustrate concepts (diagrams, videos, models). 2.1 Clarify vocabulary and symbols using visuals, connections, and supports. 3.2 Highlight big ideas, patterns, and critical features using organizers and examples.	please check student Teams
Teacher's notes/ Reminders ملاحظات المعلم والتذكير	Bring a charged devices, calculator, Physics notebook, ruler, pencil daily, participate actively, review feedback and upload required work to Toddle and Teams.		

Week # 4 الأسبوع#

Technology Integration دمج التكنولوجيا	Digital graphing or teacher-provided graph displays; calculators for gradients, areas and equations; motion videos or simulations where available; Teams for resources, feedback and submission.
Cross-curricular links الروابط بين المواد الدراسية	Mathematics: gradient, geometric area, algebra, rearranging equations, proportional reasoning and negative values. English: interpret motion descriptions and justify conclusions. UAE engineering and road safety: analyse transport data and stopping motion.
External Assessment Integration دمج التقييمات الخارجية	NGSS-style external assessment questions requiring students to analyse data, interpret multi-section graphs, calculate gradient and signed area, select kinematics equations, translate between representations and justify conclusions.
Wellbeing/ ATL Skills الصحة النفسية/ أساليب التعلم	ATL Thinking: analyse patterns, connect representations and evaluate solutions. Communication: explain graph features and justify equation choices. Self-management: organise multi-step working and check signs and units. Collaboration: critique reasoning respectfully. Wellbeing: build confidence through scaffolded practice and purposeful challenge.
Standards المعايير	HS-PS2-1: Analyze data to support the claim that Newtons second law of motion describes the mathematical relationship among the net force on a macroscopic object, its mass, and its acceleration. Supporting alignment: students analyse motion data and graphical acceleration as the kinematic foundation for later force-motion modelling.
Inquiry/ TOK Question البحث والاستقصاء أسئلة TOK	How can the same journey be represented by equations and different motion graphs, and how can we verify that the representations are consistent?
Success Criteria معايير النجاح **(Measurable)**	**Lesson 1** **・ Must: Describe the Doppler effect in terms of changes in observed frequency.** **・ Should: Explain how the observed frequency changes when a wave source approaches or moves away from an observer.** **・ Could: Apply the Doppler effect to real-life examples, such as a moving ambulance or a star moving relative to Earth.** **Lesson 2:** **・ Must: Identify nodes and antinodes in a standing-wave pattern.** **・ Should: Explain how two waves of the same frequency and similar amplitude travelling in opposite directions produce a standing wave.** **・ Could: Apply standing-wave principles to calculate wavelength or frequency in strings or air columns.** **Lesson3:** **・ Must: Identify the nucleon number \(A\), proton number \(Z\), and chemical symbol \(X\) in nuclear notation \(\,{}^{A}_{Z}X\).** **・ Should: Determine the numbers of protons, neutrons, and electrons in an atom or ion using nuclear notation, and distinguish between emission and absorption spectra.** **・ Could: Explain how discrete spectral lines provide evidence that electrons occupy quantised energy levels and relate photon emission or absorption to transitions between these levels.**
National Identity Domain مجالات الربط بالهوية الوطنية	Belonging الإنتماء Global Understanding العال م التفاهم History التاري••••خ **Area:** UAE Transportation, Road Safety, Innovation and Sustainable Infrastructure المجال Implementation: Students analyse Abu Dhabi transport motion using graphs, gradients, signed areas and kinematics equations, connecting accurate modelling to UAE road safety and smart mobility. التطبيق